

The Two Faces of Time

The Distinction Between Coordinate Time, Proper Time, and Duration

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<https://github.com/hmahaffeyges/IAM-Validation>

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Abstract

Time is two-faced. The first face, coordinate time as a redundant label on the spacetime manifold, was identified by the Machian relational framework [Barbour & Bertotti, 1982]. The second face has remained elusive. We ask whether *duration* is equally redundant, and propose that it is not. Duration is real, irreversible, and has a specific physical origin. The relational framework not only accommodates this finding, it points directly toward it.

We propose a distinction among three quantities that are often treated interchangeably: (i) coordinate time, a label on the manifold with no preferred direction and no accumulation, redundant as a fundamental concept; (ii) proper time τ , geometric path length along a worldline, reversible in principle; and (iii) duration, measured by $E(a) = \exp(1 - 1/a)$, thermodynamic, irreversible, beginning at the electroweak transition, accumulating on massive particles only.

The distinction is not terminological. Photons travel null geodesics: $d\tau = 0$. They plot on the manifold but accumulate neither proper time nor duration. Massive particles travel timelike worldlines, accumulate both τ and $E(a)$, and constitute the matter sector. This sector separation, photon sector and matter sector, follows directly from the geometry of the manifold combined with the thermodynamic cost of irreversible decoherence events on timelike worldlines.

The arrow of time is not a statistical artifact of initial conditions. It is a thermodynamic necessity: each decoherence event writes information permanently on the cosmic horizon, with each bit erased carrying a minimum thermodynamic cost of $k_B T \ln 2$ [Landauer, 1961]. The accumulated ledger is $E(a)$, monotonically increasing by construction. Duration has direction because actualization has direction. Potential moves toward act, never the reverse.

Keywords: time; duration; Machian relationism; arrow of time; electroweak symmetry breaking; gravitational decoherence; sector separation; Informational Actualization Model

1 Two Faces

Time is two-faced.

Barbour identified the first face in 1982 with the Machian relational framework: coordinate time is redundant, a label on a dimension, not a river carrying us forward [Barbour & Bertotti, 1982]. That result has shaped how foundational questions about time are posed.

The second face has remained elusive. Is duration equally redundant? We followed that question and found something unexpected. Duration is real, irreversible, and has a specific physical origin. And the relational framework Barbour built not only accommodates this finding, it points directly toward it.

2 Three Quantities Worth Distinguishing

Time appears in physical theory in at least three distinct roles that are worth distinguishing carefully. The differences between them are not merely terminological.

2.1 Coordinate time: the map

Coordinate time, t or the scale factor a , is a label on the spacetime manifold. It assigns a number to each event. It does not require anything to happen. It does not require worldlines to exist, mass to be present, or irreversible processes to occur. The manifold exists with its coordinate structure whether or not any event within it is irreversible.

Barbour's central insight [Barbour & Bertotti, 1982, Barbour, 1994] is that this coordinate has no privileged status. It is redundant as a fundamental concept. The relational framework constructs dynamics from configurations alone, without reference to an external temporal parameter. Coordinate time is the map. It is not the territory.

2.2 Proper time: path length on the map

Proper time τ is the geometric path length along a worldline:

$$\tau = \int \sqrt{-g_{\mu\nu} dx^\mu dx^\nu}. \quad (1)$$

It is still a geometric quantity, a property of the spacetime manifold measured along a particular path. A clock measures proper time. It tells you how far you have traveled on

the map.

Critically, proper time is reversible in principle. In a universe with time-reversal symmetric laws, a worldline running forward and backward in coordinate time accumulates τ in both directions. Proper time distinguishes clocks from one another but does not, by itself, distinguish past from future.

For null geodesics, photons, $d\tau = 0$ identically. Photons plot on the manifold but accumulate zero proper time. This is not an approximation. It is the definition of a null geodesic.

2.3 Duration: the journey itself

Duration is categorically different from both coordinate time and proper time. It is not a geometric quantity. It is thermodynamic.

Duration measures the accumulated irreversible acts on a worldline, the decoherence events that write classical information permanently on the cosmic horizon at Landauer cost $k_B T \ln 2$ per bit [Landauer, 1961]. It requires:

1. Massive particles on timelike worldlines ($d\tau > 0$),
2. Irreversible quantum-to-classical transitions (gravitational decoherence),
3. A preferred direction established by CP violation at the electroweak transition [Sakharov, 1967].

Before the electroweak symmetry breaking at $t \approx 10^{-12}$ s: no massive particles, no timelike worldlines, every process time-reversible. Duration cannot accumulate. $E(a) \equiv 0$.

After the electroweak transition: the matter sector exists, CP violation locks in the arrow of time at the particle physics level, and duration begins. $E(a)$ starts its 13.8-billion-year journey.

The map and the ledger are shown in Figure 1. The same worldline admits two categorically different descriptions.

3 The Second Face: Duration

3.1 The activation function

Within the Informational Actualization Model (IAM, Mahaffey 2026a), the accumulated duration since the electroweak transition is measured by the activation function:

$$E(a) = \exp\left(1 - \frac{1}{a}\right), \quad (2)$$

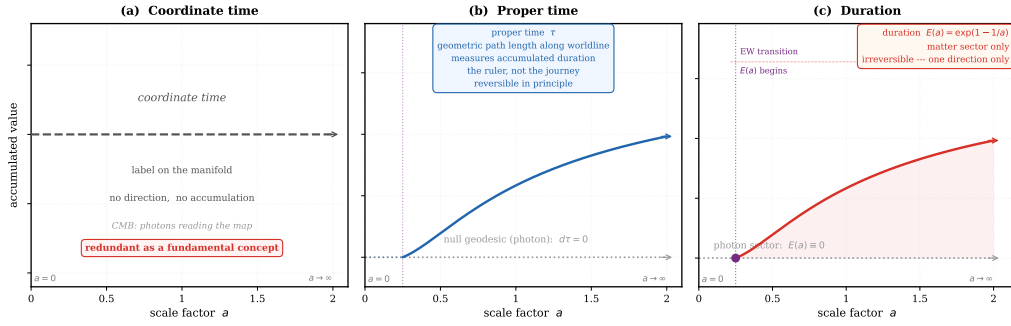


Fig. 1. Three quantities physics has conflated. (a) Coordinate time: a label on the manifold, redundant as a fundamental concept. (b) Proper time τ : geometric path length along a worldline, reversible in principle; photons accumulate none ($d\tau = 0$). (c) Duration $E(a) = \exp(1 - 1/a)$: thermodynamic, irreversible, beginning at the electroweak (EW) transition, accumulating on massive particles only. The matter sector and photon sector are separated by construction. IAM framework: Mahaffey (2026), doi:10.5281/zenodo.18702042

Figure 1: The two faces of time. Panel (a): geometric quantities — coordinate time (redundant label, Barbour & Bertotti 1982) and proper time τ (path length along world-line). Photons plot on the manifold but accumulate neither. Panel (b): duration $E(a) = \exp(1 - 1/a)$, a thermodynamic quantity categorically distinct from both. Irreversible, beginning at the electroweak transition, accumulating on massive particles only. The matter sector and photon sector are separated by construction.

where a is the cosmological scale factor normalized to unity today.

The properties of $E(a)$ follow directly from its physical interpretation:

- $E(a) \rightarrow 0$ as $a \rightarrow 0$: no duration before matter exists.
- $E(1) = 1$: normalized to today.
- $E(a) \rightarrow e$ as $a \rightarrow \infty$: duration approaches but never reaches Pure Act; the asymptote is approached, never attained.
- $dE/da > 0$ always: monotonically increasing, one direction only.

The monotonic increase is not imposed. It follows from the irreversibility of each decoherence event. Each bit written on the horizon cannot be unwritten. The ledger only accumulates. This is the arrow of time, not statistical, not initial-condition-dependent, but thermodynamically necessary.

3.2 Photons do not accumulate duration

Photons travel null geodesics. $d\tau = 0$ by construction. No proper time accumulates. No gravitational decoherence occurs; decoherence requires nonzero proper time on a timelike worldline. Therefore:

$$E(a)|_{\text{photon}} \equiv 0. \quad (3)$$

This is the physical basis for the sector split. Photons interact with the spacetime geometry; they are bent by gravity, redshifted by expansion, but they do not write in the

ledger. The modification to gravitational dynamics that $E(a)$ produces applies exclusively to the matter sector.

3.3 The Machian framework already knew this

The relational framework insists that physics be built from observable relations between things. Coordinate time fails this test, it is not directly observable. Proper time passes, clocks measure it. Duration passes with particular force: it is the only temporal quantity that distinguishes past from future by means of a physical mechanism rather than a statistical assumption.

Mach’s criterion, “time is an abstraction at which we arrive through the changes of things”, selects duration over coordinate time immediately. But not all changes are equal. Reversible changes do not advance the ledger. Only irreversible changes, only decoherence events, only Landauer costs paid, constitute duration in the sense that gives time its direction.

The relational framework points naturally toward this distinction. The physical mechanism, the electroweak transition establishing irreversibility and the Landauer cost quantifying it, connects the relational picture to thermodynamics.

4 What the Sector Split Implies

The distinction between the map and the ledger has an immediate observational consequence that may help clarify a persistent tension in cosmological observations.

The CMB measurement of H_0 uses photons emitted at last scattering, traveling null geodesics for 13.8 billion years. The entire measurement is geometric. No proper time accumulated. No duration written. It reads the map. $H_0^{\text{photon}} = 67.16 \text{ km s}^{-1} \text{ Mpc}^{-1}$ is what the manifold says about its own expansion rate [Planck Collaboration, 2020].

The local distance ladder uses Cepheids, Type Ia supernovae, and TRGB stars, massive objects on timelike worldlines, that have been accumulating duration for billions of years. The calibration is built from clocks that tick. It reads the ledger. $H_0^{\text{matter}} = 72.26 \text{ km s}^{-1} \text{ Mpc}^{-1}$ is what the matter sector says about its own expansion rate [Riess et al., 2022, Mahaffey, 2026b].

Within this framework, these are not necessarily two measurements of the same quantity. They may be measurements of two distinct quantities — one geometric, one thermodynamic, that a purely geometric treatment does not distinguish. The Hubble tension is consistent with the sector split, and the predicted values are derived from first principles with zero free parameters beyond ΛCDM .

The predicted split follows from a single coupling constant derived from the virial

theorem:

$$\beta_m = \frac{\Omega_m}{2} = 0.1575, \quad (4)$$

with zero free parameters beyond Λ CDM [Mahaffey, 2026a].

5 The Relational Framework and Shape Dynamics

Shape dynamics [Barbour, 2011, Barbour, Koslowski & Mercati, 2014] replaces four-dimensional spacetime refoliation invariance with three-dimensional conformal invariance. The universe is described by the evolution of shape, the relational configuration of matter, without reference to absolute scale or absolute time.

IAM is compatible with this framework at every level. IAM does not modify the background geometry. It does not alter the Friedmann equations. It operates at the perturbation level, adding a single Hubble friction term to the matter perturbation equations:

$$\ddot{\delta}_m + 2H\dot{\delta}_m - \frac{3}{2}\Omega_m H^2 \mu(a) \delta_m = 0, \quad (5)$$

where

$$\mu(a) = \frac{H_{\Lambda\text{CDM}}^2}{H_{\Lambda\text{CDM}}^2 + \beta_m E(a)} < 1. \quad (6)$$

The geometric arena, Platonia, shape space, the relational configuration, is untouched. Duration is what happens on the worldlines *within* that arena. It does not modify the arena itself.

Shape dynamics asks: what is the universe's shape? IAM asks: what has the universe written in the process of becoming its current shape? The two questions are complementary, not competing.

Platonia provides the map of all possible configurations. IAM reveals what happens at the boundary when one is chosen. Each event written into history and onto the map, where potential becomes actual, one decoherence event at a time.

6 An Open Question

If duration begins at the electroweak transition at the moment the Higgs field, undergoing spontaneous symmetry breaking through a quantum fluctuation, selected a direction in field space irreversibly: was that selection itself the first decoherence event? The first information written on the horizon, the first Landauer cost paid, the first moment $E(a)$ could step away from zero? This remains an open question [Mahaffey, 2026c].

7 Repository

The complete thermodynamic derivation is developed in the IAM theory paper [Mahaffey, 2026a]. The full collection of papers addressing observational consequences, foundational extensions, open problems, and future predictions is available in the documents folder of the repository at <https://github.com/hmahaffeyges/IAM-Validation>. Follow whichever thread interests you.

Acknowledgments

The debt to Jacobson [1995], Cai and Kim [2005], and Barbour and Bertotti [1982] is explicit throughout. This work builds directly on their foundational contributions.

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